Data Science

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Homework
in preparation of

Lecture 7
–
Feature Engineering

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Feature Engineering

Feature engineering is the process of using domain knowledge to extract features from raw data in order to improve the performance of machine learning models. We will be looking at the most important Feature Engineering techniques in the upcoming lecture.

In preparation, please explore and clean the dataset, which we will be using in the lecture to practise feature engineering:

- Complete the notebook ”h07a...” - (expected time: **120min**)

In addition, we please review/familiarize with the material on the following slide covering

- Features (Categorical, Ordinal, Boolean, Numeric)
- Clustering, in particular the k-means Clustering Algorithm
- PCA

Important kinds of Features

- Quantitative = Numerical Features
  - Discrete Values: only certain values are permitted, e.g., whole numbers
    Example Feature: number of employees (may be 1 or 2 or 15, but never 3.75)
  - Continuous Values: values fall anywhere, usually within a certain range
    Example Feature: Length of flower petal (may be anywhere in the range (0, ∞))
- Qualitative = Categorical Feature
  - Nominal Values: categories with no order
    Example Feature: hair color (may be blond, black, red, …)
  - Ordinal Values: categories with a given order (or hierarchy)
    Example Feature: wine quality (may be low, medium, high, where low < medium < high)
    or job titel (may be individual_contributer < manager < director)
  - Special Case: Binary=Boolean Values have exactly two different values (may be ordered or not)
- There are many more kinds of features like times, dates, intervals, …

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Review of Linear Regression and Regularization

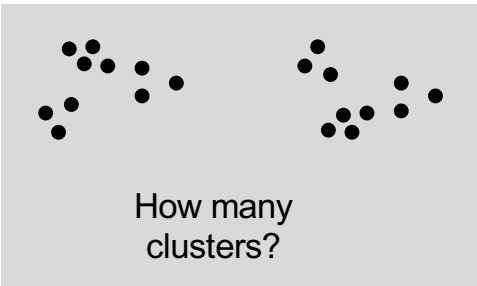
- We will be referring to linear Regression in the upcoming class frequently – both to the **Simple Linear Regression** as well as to their **regularized** Version (**Lasso** and **Ridge** Regression).
- If you haven't done so already, please review the material from the book "Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani: An Introduction to Statistical Learning (with appl. in Python), Springer, 1st Edition, 2023." (available as ebook via the TH library):
  - Read Chapter 1
  - Read Chapter 2
  - Read Chapter 3 intro and 3.1, 3.2
  - Read Chapter 6 intro and 6.2

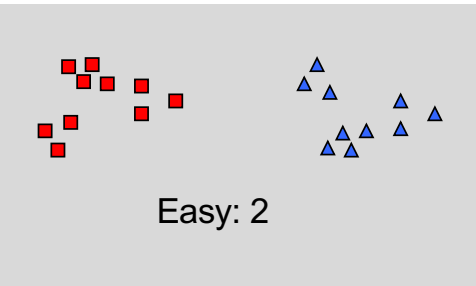
Clustering

- Cluster analysis or clustering is the task of grouping a set of objects in such a way that objects in the same group (called a cluster) are more similar (in some sense) to each other than to those in other groups (clusters). It is a main task of exploratory data analysis, and a common technique for statistical data analysis.

- Cluster analysis itself is not one specific algorithm, but the general task to be solved. It can be achieved by various algorithms that differ significantly in their understanding of what constitutes a cluster and how to efficiently find them. Popular notions of clusters include groups with small distances between cluster members, dense areas of the data space, intervals or particular statistical distributions. Clustering can therefore be formulated as a multi-objective optimization problem. The appropriate clustering algorithm and parameter settings (including parameters such as the distance function to use, a density threshold or the number of expected clusters) depend on the individual data set and intended use of the results. Cluster analysis as such is not an automatic task, but an iterative process of knowledge discovery or interactive multi-objective optimization that involves trial and failure. It is often necessary to modify data preprocessing and model parameters until the result achieves the desired properties. [https://en.wikipedia.org/wiki/Cluster_analysis]

- “Cluster“ and clustering are to some extent ill-defined!









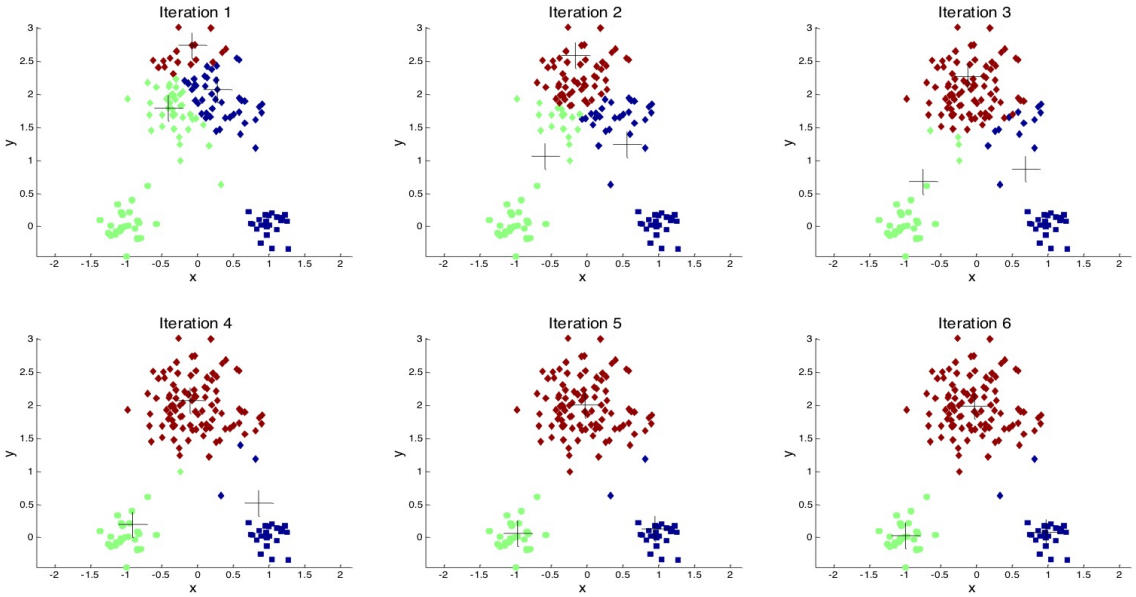
k-means Clustering

- Idea: Pick random initial Cluster Centers (Centroids) and iterativly move these to improve the Clustering
  - Data: X = (x^p_1, ..., x^p_d) in Euclidean space with L_2-norm
  - Centroid element is mean value of current assignments
  - *k* is given apriori (must be chosen)
  - See e.g., https://en.wikipedia.org/wiki/K-means_clustering

Algorithm: k-means(data, dist, k)

1: select *k* values as initial centroids c(k)
2: REPEAT
3: Assign each data point to cluster based on distance to centroid
4: Recompute centroids
5: UNTIL change in centroids less than ε

- *Example:*



PCA (1)

- Principal component analysis (PCA) is a popular technique for analysing large datasets containing a high number of dimensions/features per observation, increasing the interpretability of data while preserving the maximum amount of information, and enabling the visualization of multidimensional data. Formally, PCA is a statistical technique for reducing the dimensionality of a dataset. This is accomplished by linearly transforming the data into a new coordinate system where (most of) the variation in the data can be described with fewer dimensions than the initial data. Many studies use the first two principal components in order to plot the data in two dimensions and to visually identify clusters of closely related data points. Principal component analysis has applications in many fields such as population genetics, microbiome studies, and atmospheric science.

- In data analysis, the first principal component of a set of p variables, presumed to be jointly normally distributed, is the derived variable formed as a linear combination of the original variables that explains the most variance. The second principal component explains the most variance in what is left once the effect of the first component is removed, and we may proceed through p iterations until all the variance is explained. PCA is most commonly used when many of the variables are highly correlated with each other, and it is desirable to reduce their number to an independent set.

- PCA is used in exploratory data analysis and for making predictive models. It is commonly used for dimensionality reduction by projecting each data point onto only the first few principal components to obtain lower-dimensional data while preserving as much of the data's variation as possible. The first principal component can equivalently be defined as a direction that maximizes the variance of the projected data. The i-th principal component can be taken as a direction orthogonal to the first i-1 principal components that maximizes the variance of the projected data.

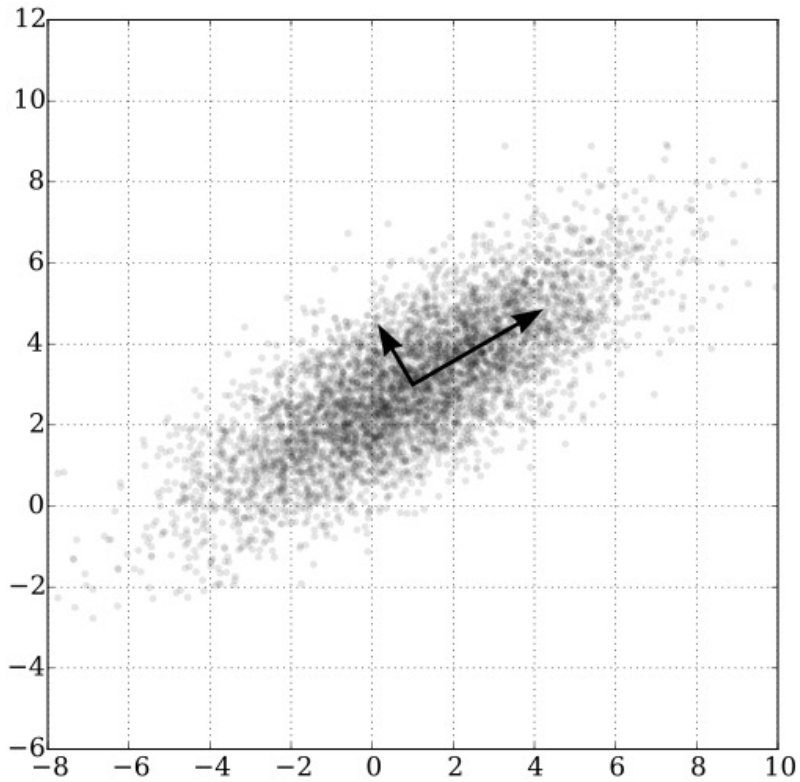
PCA (2)

Intution

- PCA can be thought of as fitting a p-dimensional ellipsoid to the data, where each axis of the ellipsoid represents a principal component. If some axis of the ellipsoid is small, then the variance along that axis is also small.

- To find the axes of the ellipsoid, we must first center the values of each variable in the dataset on 0 by subtracting the mean of the variable's observed values from each of those values. These transformed values are used instead of the original observed values for each of the variables. Then, we compute the covariance matrix of the data and calculate the eigenvalues and corresponding eigenvectors of this covariance matrix. Then we must normalize each of the orthogonal eigenvectors to turn them into unit vectors. Once this is done, each of the mutually-orthogonal unit eigenvectors can be interpreted as an axis of the ellipsoid fitted to the data. This choice of basis will transform the covariance matrix into a diagonalized form, in which the diagonal elements represent the variance of each axis. The proportion of the variance that each eigenvector represents can be calculated by dividing the eigenvalue corresponding to that eigenvector by the sum of all eigenvalues.

Source and more Details:
https://en.wikipedia.org/wiki/Principal_component_analysis



Source: Nicoguaro, https://commons.wikimedia.org/wiki/File:GaussianScatterPCA.svg

